Subject: Submission of "Optimizing Semantic Retrieval for Bengali: A Comparative Analysis of Monolingual and Multilingual Embeddings with Matryoshka Representation Learning"

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled "**Optimizing Semantic Retrieval for Bengali: A Comparative Analysis of Monolingual and Multilingual Embeddings with Matryoshka Representation Learning**" for consideration in *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*.

Retrieval-Augmented Generation (RAG) is transforming NLP, yet its effectiveness in low-resource languages like Bengali is often hindered by suboptimal retrieval components. In this study, we address this gap by conducting a rigorous comparative analysis of three distinct embedding architectures: a monolingual Bengali model, a distilled multilingual model, and a paraphrase-centric model (MPNet).

Our work offers two primary contributions relevant to the TALLIP audience:

1. **Benchmarking:** We demonstrate that while monolingual models are popular, a fine-tuned multilingual MPNet architecture statistically outperforms them (NDCG@10 of 0.8114).
2. **Efficiency for Low-Resource Deployment:** We are among the first to apply Matryoshka Representation Learning (MRL) to Bengali. We show that our model retains 96% of its performance at just 128 dimensions, reducing storage costs by 83%. This is crucial for deploying semantic search in resource-constrained environments typical of the developing world.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. We believe this work aligns perfectly with TALLIP's mission to advance information processing technologies for Asian and low-resource languages.

Thank you for your consideration.

Sincerely,

Koshik Debanath
Department of Computer Science and Engineering
Rajshahi University of Engineering and Technology (RUET)